ModelIt!

Student Activity Pack

Why Can We See Through Some Things?

The Science of Light and Materials

Grade 1 | NGSS 1-PS4-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw something you can see through and something you cannot see through.

2 Circle the things that light can pass through: a window, a brick wall, a clear water bottle, a book, plastic wrap.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Material Type | Light Passing Through

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Clear Plastic Cup

Material Type set to transparent

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cardboard Cup

Material Type set to opaque

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why can you see through a window but not through a wall? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Flashlight Viewer

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YOUR ENGINEERING MISSION

Build a tube viewer that uses different materials to change how much light gets through, like a fun light filter.

THE SCENARIO

A toy company wants you to design a cool flashlight viewer! When you look through it, you should be able to change what you see by switching the material on the end. Use what you learned about transparent, translucent, and opaque materials.

Engineering Design Thinking

1. WHAT WILL YOU USE FOR THE TUBE PART OF YOUR VIEWER?
2. WHAT THREE DIFFERENT MATERIALS WILL YOU TEST ON THE END?
3. HOW WILL YOU ATTACH THE MATERIALS SO YOU CAN SWITCH THEM?
4. WHICH MATERIAL LETS YOU SEE THE MOST? WHICH LETS YOU SEE THE LEAST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Window Designers and Architects choose the right materials so buildings let in the perfect amount of light. They earn $50,000-$90,000/year.